

Solving Recurrence Relations: The Substitution Method

Introduction

In the analysis of recursive algorithms, recurrence relations describe the running time. The **Substitution Method** (also called the Guess and Prove method) is used to find an asymptotic upper and lower bound by:

1. **Guessing** the solution.
2. **Proving** it using mathematical induction.

Steps in the Substitution Method

1. Guess the form of the solution (often using experience or by expanding the recurrence a few times).
2. Prove the guess by induction:
 - **Inductive Hypothesis:** Assume the solution holds for all values smaller than n .
 - **Inductive Step:** Substitute the hypothesis into the recurrence and show it satisfies the bound.
 - Adjust constants if necessary.

Example: Solve $T(n) = 2T(n/2) + n$

Step 1: Guess. Based on the recurrence, we guess:

$$T(n) = O(n \log n)$$

Step 2: Inductive Hypothesis. Assume for all $k < n$, we have:

$$T(k) \leq c \cdot k \log k$$

Step 3: Inductive Step.

$$\begin{aligned}T(n) &= 2T(n/2) + n \\&\leq 2 \cdot c \cdot \frac{n}{2} \log\left(\frac{n}{2}\right) + n \\&= cn \log\left(\frac{n}{2}\right) + n \\&= cn(\log n - \log 2) + n \\&= cn \log n - cn + n \\&= cn \log n + n(1 - c)\end{aligned}$$

To ensure $T(n) \leq cn \log n$, we require:

$$1 - c \leq 0 \Rightarrow c \geq 1$$

So the upper bound holds if $c \geq 1$.

Conclusion:

$$T(n) = O(n \log n)$$

Tips for Students

- Use iteration first to guess the solution.
- Don't worry about exact constants; focus on the form of the solution.
- You may also need to prove a lower bound $T(n) = \Omega(f(n))$ to establish $T(n) = \Theta(f(n))$.

Other Common Recurrences

- $T(n) = T(n-1) + c \Rightarrow T(n) = \Theta(n)$
- $T(n) = 2T(n/2) + n \Rightarrow T(n) = \Theta(n \log n)$
- $T(n) = T(n/2) + 1 \Rightarrow T(n) = \Theta(\log n)$